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ENERGY COOP · EMBASSY

Energy communities

CleantechHUB as a partner against energy poverty: public data, O&M cooperatives, and Colombia's cleantech ecosystem.

Albert Kobus · 11 Nov 2026 · cleantechhub.net



Photo: Tierra Grata · rural communities, Colombia



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CleantechHUB

Climate innovation hub in LATAM: cleantech startups, corporates, investors, and government.

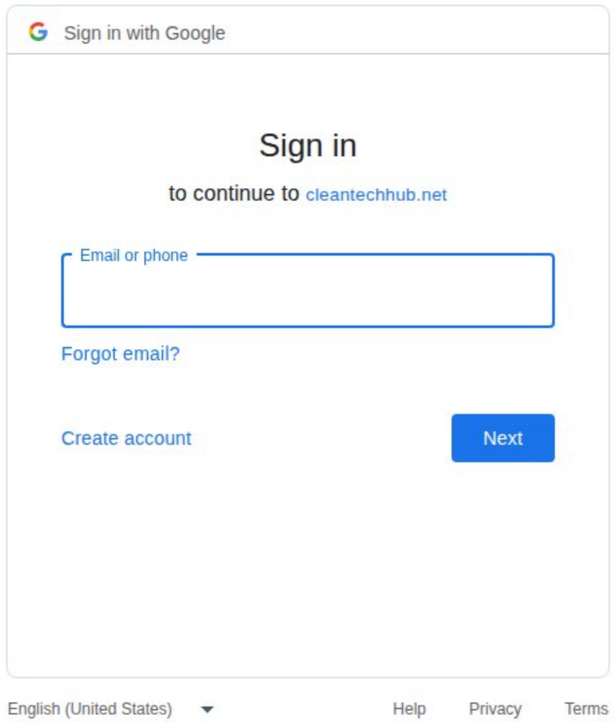
2019
Founded (cleantechhub.net)

5
Countries cited: CO, PE, SV, GT, CR

3
Platforms: web · wiki · Nexus

LATAM
Regional focus (schema.org)

Source: cleantechhub.net JSON-LD Organization, Oct 2026.



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ClimateLaunchpad CLP26

National Organizer for the world's largest green business ideas competition — pipeline to Nexus.

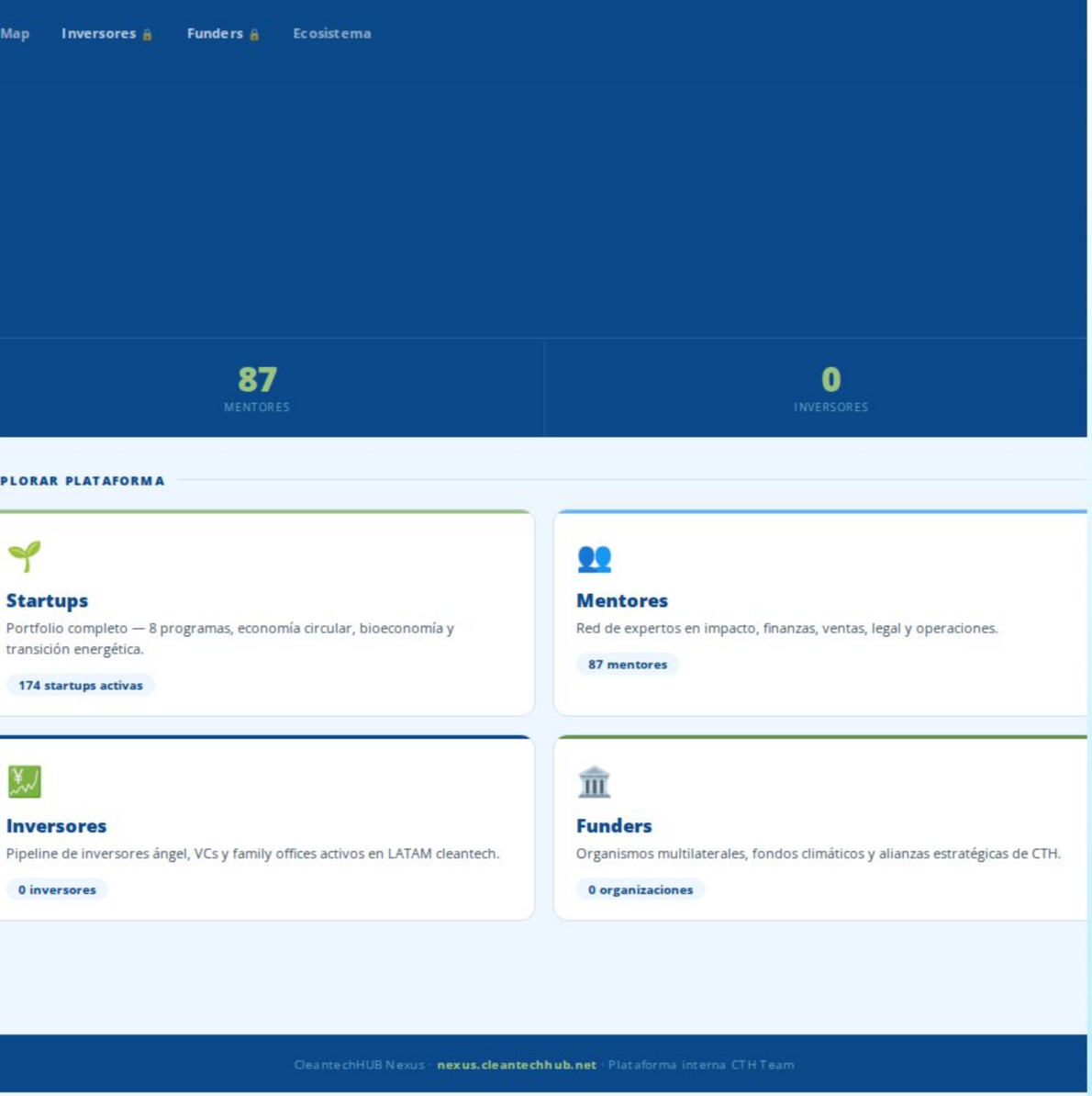
5
2026 campaign countries: CO, PE, SV, GT, CR

CLP
National Organizer · CleantechHUB

6–8
Week call window (clp26-brand)

decks.cleantechhub.net/cth-clp26 | ClimateLaunchpad global

Source: skills/clp26-brand/SKILL.md · CLAUDE.md.



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Nexus

Bilingual cleantech startup portfolio — profiles at nexus.cleantechhub.net.

5
Energy Coop startups (brochure-v2)

EN·ES
Bilingual startup profiles

Sunco · SEOS · Soluna · Tierra Grata · TuWatt

Source: skills/nexus-onepager · brochure-v2.



CASOS DOCUMENTADOS

Lo que ya funciona, y por qué

Nueve cooperativas y marcos regulatorios europeos con cuarenta años de recorrido, y los proyectos comunitarios que Colombia ya tiene en operación. Con sus cifras, sus vacíos y sus fallas.

9	31	3
casos europeos analizados	proyectos colombianos documentados	países de referencia

DÓNDE ESTÁN

Mapa de los casos

Seis de los nueve casos europeos tienen un lugar declarado en la fuente. Los otros tres no son organizaciones sino marcos nacionales, y no se ubican en el mapa.

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Energy Coop

Energy cooperatives: social layer and O&M where CAPEX and an implementer exist — no MW funds.

15.4%
National IMPE 2024 (~8.1M people)

2.9M
People to reach 2030 target

898
kWp Sunco · DISPAC La Guajira[9]

1,947
Tierra Grata energy solutions[11]

Source: brochure-v2 · IMPE 2025 · energiacoop.cleantechhub.net/casos/

REIN

Resilience · Entrepreneurship · Innovation · Nature

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REIN Hubs

LATAM network of physical hubs — cleantech activation, incubation, and acceleration with PVBLIC Foundation.

5+
Active LATAM hubs (REIN deck)

CLP
Global ClimateLaunchpad pathway

CTH
Delivery partner · no $ figures here

PVBLIC Foundation | Kintec Academy

Source: decks/rein-academic-city-ghana — AGENTS.md (no REIN dollar amounts).

P4G

Partnership for Green Growth · MubOn archive

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PRIOR

P4G / MubOn

Competitive P4G proposals and MubOn packs in CTH archive — strict co-finance rules, no invented figures.

50/50
Typical P4G co-financing (cth-grant)

Grant
UNDP graph · P4G/MubOn (Drive)

P4G | MubOn | UNDP context

Source: skills/cth-grant/SKILL.md · docs/grant-graph/spec.md.

The gap and OPEX

15.4%
IMPE 2024 · ~8.1M

9.5%
2030 target · ~2.9M

75.9%
Rural ICEE 2024

63.1%
IMPE La Guajira

When do we close the 2.9M-person gap?

Observed pace ~300k/year



IMPE target ~490k/year × 6 years



Contraloría
Microgrid O&M not remunerated.

Amazonas
~$30,000M royalties in idle PV.

CREG
Abandoned rural PV without management.

CTH: *bankable* social layer and OPEX with committed CAPEX and implementer.

PROPOSAL

Cooperative + ecosystem

Community O&M · cooperative governance · IMPE dimension KPIs (does not replace ICEE).

SN	SE	SL	TG	TW
Sunco	SEOS	Soluna	Tierra Grata	TuWatt
898 kWp[9]	+75 SOLvers[10]	ZNI	1,947[11]	Coops

Candidate: Corozal, Sucre (IMPE 43.1%) — not a PASO energy project.

Data lab (illustrative)



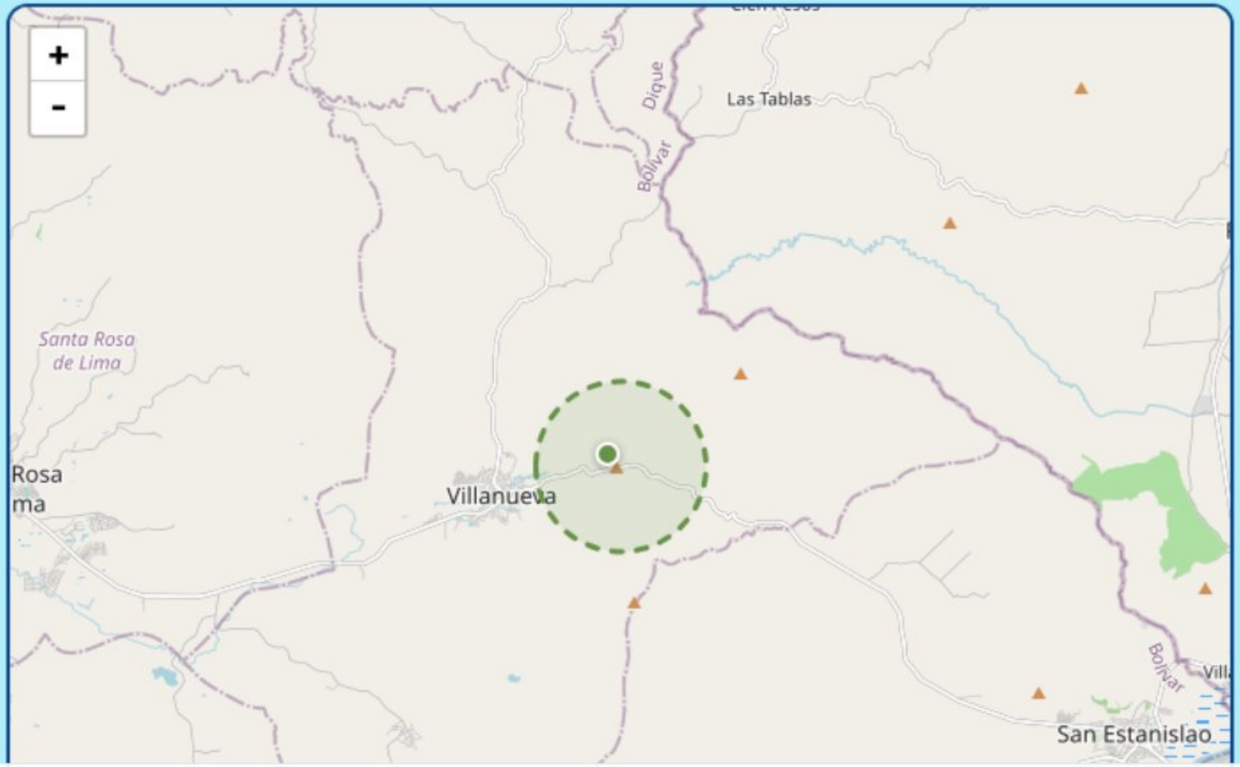
~1,000 Public cross-cut
100 Scoring
20 Validation
10 Co-op wave

Paiva: from committee to cooperative

Community solar energy in vereda Paiva, Bolívar (~450 inhabitants), IE Chiricoco. NOT FENOGE Map location **approximate**.

National energy poverty (IMPE, index on persons): 15.4% · ~8.1 M people (2024)
Informe IMPE 2025 (PDF) · Nota IMPE, nov 2025

Rural electricity coverage 75.92% vs national 93.12% (UPME ICEE 2024)
Boletín ICEE 2024 (UPME) · UPME, noticia 2024



PUBLIC SOURCE

Community solar project with local committee and IE Chiricoco (linked public sources).

Noticia Paiva (2025) · Guía del Gas

CTH DESIGN PROPOSAL

Guided transition: community committee → energy cooperative with ~30% participation in the community share class (CTH proposal, not a contractual figure).

Community committee
Grassroots organization

→

Energy cooperative
Legal personality

Transition shown

- General assembly — supreme social body
- Board of directors — oversight and strategy
- Operator — technical O&M of the asset
- Energy-sector ally — know-how (role per local agreement)
- Community share class — ~30% CTH proposal in capital structure

La Guajira at scale

Public projects in La Guajira (FENOGE and related processes). Pins without official coordinates: *approximate location*.
Verification: `docs/energy-coop/demo-cth-brand/SOURCES.md`.

IMPE La Guajira 63.1% (2024)
Informe IMPE 2025 (PDF) · PARTIAL — 1ST DEPT. SOURCE

ICEE La Guajira 54.28% (UPME 2024)
Boletín ICEE 2024 · UPME, noticia

Cooking with firewood/charcoal/waste: 56.6% of people in energy poverty (IMPE 2025)
Informe IMPE 2025 (PDF) · Resumen ejecutivo IMPE 2025

0	0	0
Projects on map	Verified kWp (sum)	Announced MW (utility-scale)

View La Guajira at scale



Data Lab — prioritization funnel

How the Energy Coop pipeline narrows candidates before PASO. Public policy figures and documented cross-checks from `docs/energy-coop/`; wave counts (100 → 20 → 10) are EXAMPLE DATA illustrating the flow.

- **Public universe (reference)**
  National target of **500 energy communities 2024–2025**, 40% ZNI (FENOGE). In La Guajira, IPSE reports **148 EC** and **1,409 families** under Miichi Ka'i (Dec 2024).
  Sources: `energy-poverty-kpis-2026-11.md` · FENOGE · IPSE
- **IMPE / ICEE / IPEM cross-check**
  Desk layer: departmental IMPE incidence (e.g. La Guajira 63.1%, Chocó 52.8%), rural vs urban ICEE, household IPEM (different index). Does not replace field visits.
  Cross-check: `energy-poverty-kpis-2026-11.md` · IMPE 2025 PDF
- **Solar resource layer (screening)**
  Google Solar BASE where coverage exists (e.g. Corozal, Uribia); NASA POWER / Global Solar Atlas where not (Quibdó, Bahía Solano). Roof signal, not ground truth.
  `solar-api-coverage-2026-11.md` · `solar-api-integration-plan.md`

EXAMPLE DATA

100

Candidates after public cross-check + preliminary scoring

↓

20

Prioritized after technical and governance validation

PASO — productive community candidates

Territories in dialogue for productive cooperative models. Locations and candidate status only.

CANDIDATE COMMUNITY

Corozal, Sucre

Municipal seat — reference point with Google Solar BASE coverage in map check (9.31778, -75.29583).
`solar-api-coverage-2026-11.md`

Productive communities in dialogue (territorial name only):

CANDIDATE COMMUNITY
Fonseca, La Guajira

CANDIDATE COMMUNITY
Puerto Asís / Guzmán, Putumayo

CANDIDATE COMMUNITY
La Macarena, Meta

CANDIDATE COMMUNITY
Istmina, Chocó

CANDIDATE COMMUNITY
San José del Guaviare, Guaviare

CANDIDATE COMMUNITY
Miranda, Cauca

CANDIDATE COMMUNITY
Arauca, Arauca

Your cooperative in 3 clicks

Illustrative CEOF flow (cooperative structure + investors + operator). CTH DESIGN PROPOSAL

1. Choose a community — Select —

2. System size

10 kWp | 25 kWp | 50 kWp | 100 kWp

Create cooperative

2030 target + local talent

National IMPE scenario. Index on **persons**. Sources: `energy-poverty-kpis-2026-11.md`.

IMPE 2030 target: 15.4% (8.11 M people, 2024) → 9.5% (5.17 M) in 2030 · −2.9 M people
Informe IMPE 2025, p. 33 (PDF) · Nota IMPE, nov 2025

Pace gap (cited arithmetic)

People who must leave energy poverty ≈ **2.9 M** (8.11 M − 5.17 M; IMPE 2025 p. 33).

Observed pace ~300,000 people/year

`2,900,000 ÷ 300,000 ≈ 9.7 years from 2024`

~2034

IMPE 2025 p. 12 (~300 k/year) · IMPE 2025 report (PDF)

Target scenario pace ~490,000 people/year × 6 years

`490,000 × 6 = 2,940,000 ≈ 2.9 M`

2030

IMPE 2025 p. 33 (≥1.0 p.p./year incidence)

Not a CAPEX problem, it is OPEX

- Contraloría: no methodologies to remunerate operation, microgrid administration and SFVI → maintenance and monitoring failures. El Colombiano
- Amazonas (~$30,000 million in royalties): abandoned photovoltaic project; damaged batteries — "critical state" due to sustainability. Seguimiento.co / Procuraduría
- CREG (GEN-Renovables): thousands of rural PV systems without management → "often abandoned"; missing regulation, certified installers and O&M. Circular 005-13 informe final · Concepto CREG 4022/2019 (AOM)

Local talent — documented links

Escuela Colombiana de Ingeniería Julio Garavito: Luis Carlos Sarmiento Angulo Chair — energy sector axis; academic challenge integrating

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Live demo

https://energiacoop.cleantechhub.net/demo/

← → · Space · CleantechHUB branding

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